Shubham Kumar



Technical Lead | Core Banking & Payments | AI-Assisted Engineering (GitHub Copilot, Kiro, Agents, MCP)

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PROFESSIONAL SUMMARY

- **Technical Lead with 9+ years** in **Banking and Insurance** IT, specialising in **core banking payments**.
- Lead **CBS R&D delivery** at **Sopra Banking Software** for European banks **ARGENTA** and **CRELAN**.
- Delivered **SEPA 2023** and **Instant Payment Regulation**; expert in **SWIFT / TARGET2** and **ISO 20022** (pain, pacs, camt) messaging.
- **AI-assisted engineering:** hands-on with **GitHub Copilot (agent mode)**, **Kiro (agent & spec mode)**, **custom AI agents**, **MCP integrations** and **prompt / steering design** for faster root-cause analysis, code review and test design on a large legacy COBOL codebase.
- Deep **COBOL (Mainframe & Micro Focus)** expertise with full-stack awareness of **Oracle, Java and Unix/Linux** layers.
- Strong **root-cause analysis**, **solution design** and direct **onshore client collaboration**.
- Earlier delivered insurance and retail banking systems for **CNA, ERIE and Santander**.

AI & AUTOMATION SKILLS

- **AI coding assistants:** daily use of GitHub Copilot (agent mode) and **Kiro (agent mode and spec mode)** for analysis, code changes and review on a large COBOL / Oracle / Java codebase; validate AI output against reference implementations rather than accepting it blindly.
- **Agentic workflows:** use and evaluate multi-agent pipelines (RCA analysis agent, independent challenger agent, test-case suggester) for ticket investigation and root-cause reports.
- **Prompt & context engineering:** maintain custom instructions, steering documents, skills and policy files (evidence-backed findings, unknown-marking, search-effort limits) that keep AI answers grounded and auditable.
- **MCP integrations:** connect AI assistants to Jira, GitLab and Oracle (SQLcl) through Model Context Protocol servers to fetch tickets, review merge requests and query data.
- **AI-assisted code review & QA:** generate structured MR review reports, catch defects in AI-proposed code (macro misuse, missing regression branches, invalid test scenarios) and give feedback to tool teams.
- **Spec-driven development & TDD:** requirements, design and task breakdown using **Kiro spec mode**, test-first implementation, and AI-assisted documentation.

CORE SKILLS

Languages & Platforms	COBOL (Mainframe & Micro Focus), JCL, VSAM, DB2, CICS, SQL (Oracle), Shell scripting, Unix/Linux, z/OS; working knowledge of Java (DAO beans, B2O web services)
Payments & Messaging	SEPA CT / SDD / CT Inst, Instant Payments, SWIFT, TARGET2, SWIFT gpi tracker, R-transactions, Cards; ISO 20022 pain.001/002, pacs.002/004/008, camt.054/106; XSD validation; Isabel companion files
Core Banking	Core banking platform (CBS) with GTFrame, parameter tables, alerts, accounting, screens and transaction models; CBS 3.x to 4.x patch redevelopment
Tools & Process	Git, GitLab (merge request reviews), JIRA, Endevor, File-AID, Control-M, CA-7; Agile (Scrum) and Waterfall; GitHub Copilot, Kiro, MCP, VS Code
Engineering Practices	Requirement analysis, solution design, root-cause analysis, code review, unit/regression testing, release and production support, regulatory compliance delivery

PROFESSIONAL EXPERIENCE

Technical Lead (CBS R&D)

Apr 2022 - Present

Sopra Banking Software Pvt. Ltd., Noida

- Lead core banking payment modules and mentor the team on design, code quality and merge request reviews.
- Delivered regulatory initiatives: **SEPA 2023** rulebook changes and **Instant Payment Regulation**, including outgoing R-transaction handling for payments created before the Nov 2026 deadline.
- Built the **pain.002 status report and companion file** generation for Isabel (Instant Payments): SWIFT header, message IDs, XSD-compliant structure and sender DN, resolving repeated file rejections.
- Fixed complex production defects across SEPA CT Inst, SWIFT and TARGET2 flows: batch creation from pain.001, value-date and accounting errors, pacs.002 instructing agent, and IPCE reachability checks.

- Extended payment alerts with payment-initiator data across SEPA, T2/SWIFT and standing orders, including parameter-driven activation (DM1681).
- Implemented GPI return-reason mapping for SWIFT/TARGET2 tracker messages and camt.054 / camt.106 handling.
- Redeveloped critical 3.x patches on the 4.x release line and maintained country/IBAN reference data (e.g. IBAN Registry updates).
- Worked across COBOL, Oracle DDL/metadata and Java DAO layers; coordinated with onshore stakeholders for requirement clarification; tracked delivery in JIRA and Git.
- Introduced AI-assisted workflows for the team: Copilot-driven RCA and MR review reports, steering/policy files for consistent AI output, and MCP-connected Jira/GitLab access.

Senior Software Engineer *May 2021 - Apr 2022*

Coforge India Pvt. Ltd., Noida (Santander Bank project)

- Supported banking applications; monitored daily batch cycles and Control-M schedules and resolved production abends.
- Analysed and fixed production incidents and processing errors; prepared and executed test JCLs from business requirements.
- Authored analysis and design documents.

Associate Consultant *Aug 2017 - Mar 2021*

Capgemini India Pvt. Ltd., Hyderabad (CNA Insurance, ERIE Insurance)

- Analysed requirements and delivered enhancements in COBOL, JCL, VSAM and DB2; managed deployments and versioning with Endevor.
- Prepared detailed design documents, supported BNR activities, and handled unit testing, defect fixing and production support.
- Coordinated with downstream teams and business users for release readiness.

EDUCATION

- B.Tech - 9.3 CGPA (2017) | HSC - 76% (2013) | SSC - 8.8 CGPA (2011)

ACHIEVEMENTS & ACTIVITIES

- Participant, Capgemini Technical Festival (Tech-Fiesta).
- Winner, Fashion Show, Capgemini Mahotsav 2019; Winner, Business Unit Level Cricket Tournament.